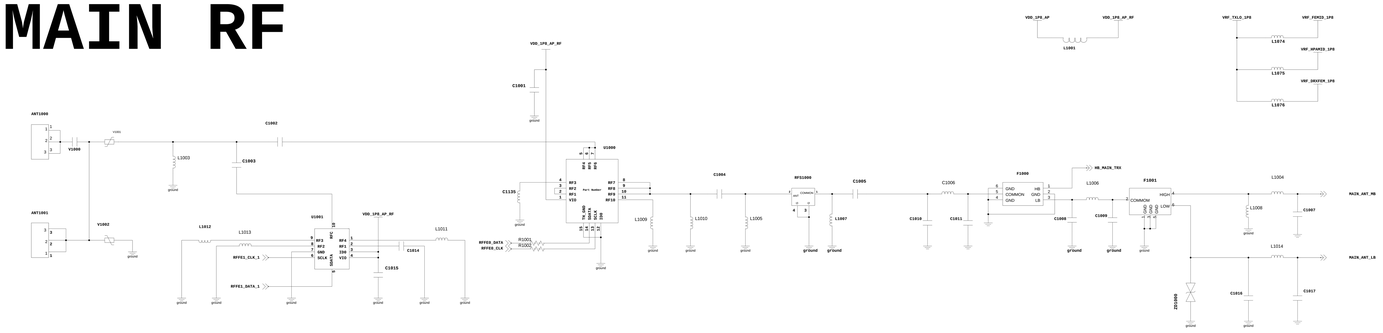
Tab_A4_XL IREPAIR

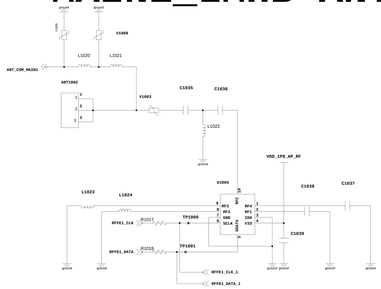
2020.02.13

	REFERENCE
SHEET01 : MAIN RF FRONT END	1000
SHEET02 : SUB RF FRONT END, TRCV, APT	2000
SHEET03 : CONNECTIVITY, SENSORS	3000
SHEET04 : AP(RAMEN)_1	4000
SHEET05 : AP(RAMEN)_2, UFS	5000
SHEET06 : PMIC(Main,IF), KEY, BATTERY	6000
SHEET07 : AUDIO, SIM/SD, SHIELD CAN, IF CON	7000
SHEET08 : DISPLAY, WACOM, CAMERA	8000
SHEET09 : IF PCB, SUB ANT	9000
SHEET10 : WIFI PCB	10000

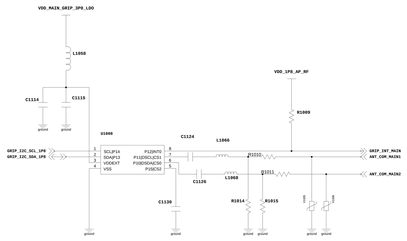
MAIN RF



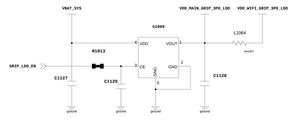
MAIN1_LMHB ANT



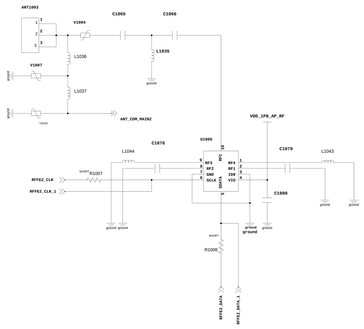
MAIN1_LMHB ANT X-GND



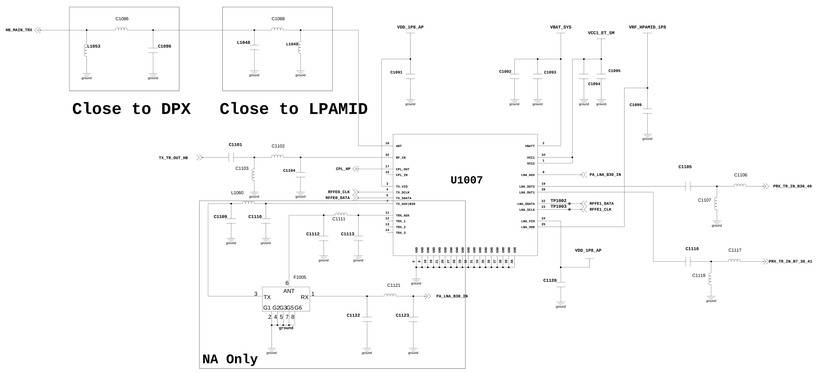
MAIN GRIP SENSOR



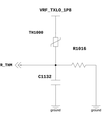
MAIN GRIP LDO



MAIN2_OPTION ANT

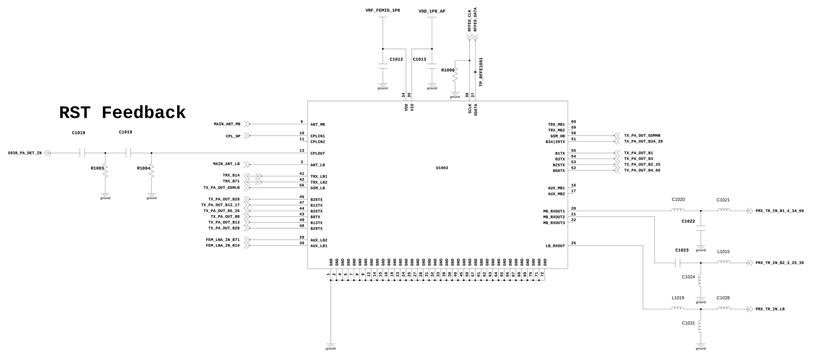


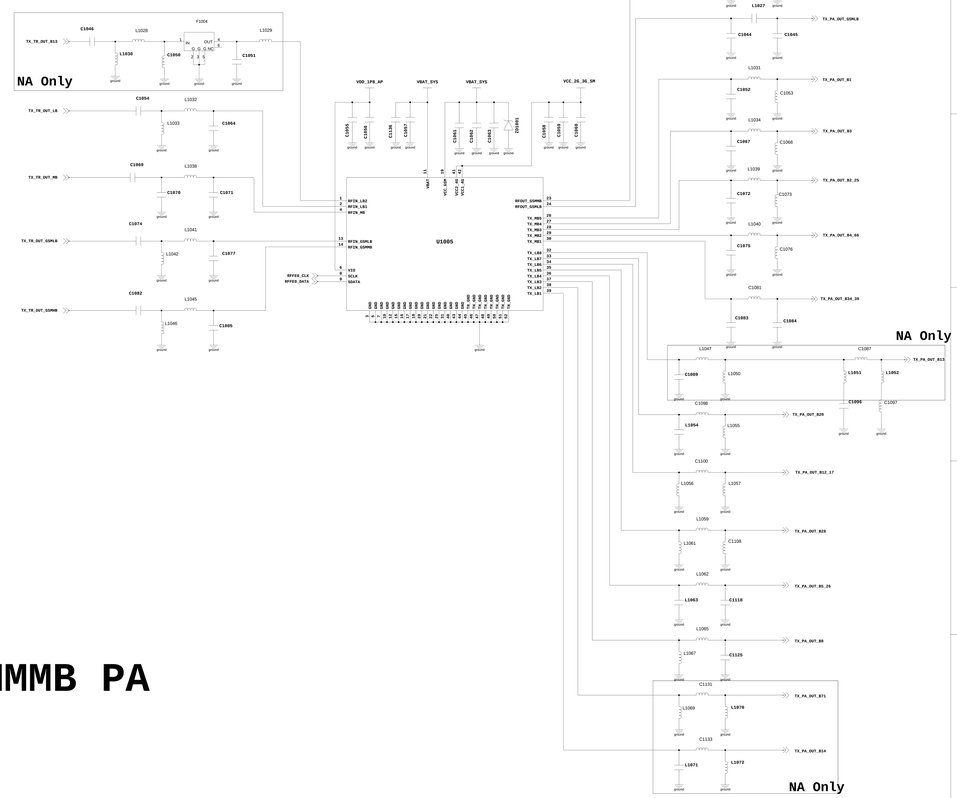
HB PAMid



THERMISTER

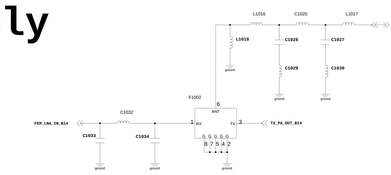
MAIN CA FEMID



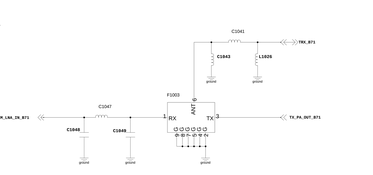


MMMB PA

NA Only



B14 DPX

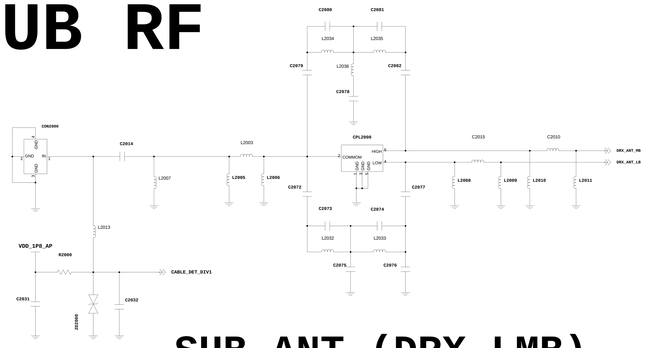


B71 DPX

TITLE	10_SM_A505GT_LA_4_64_REV0		
SIZE	A1	PCM NUMBER	Rev
SHEET	2 of 8	Drawing Date	
AUTHOR	User Name		

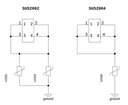
DO NOT DISCLOSE TO OR DUPLICATE FOR OTHERS
EXCEPT AS AUTHORIZED BY SAMSUNG.

SUB RF



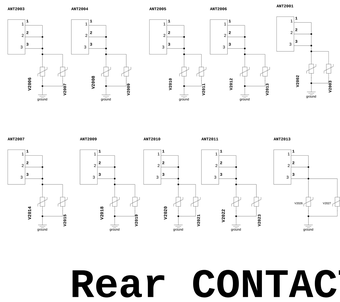
SUB ANT (DRX_LMB)

SUB ANT (DRX_HB)



SCREW SUS

Cable CLIP



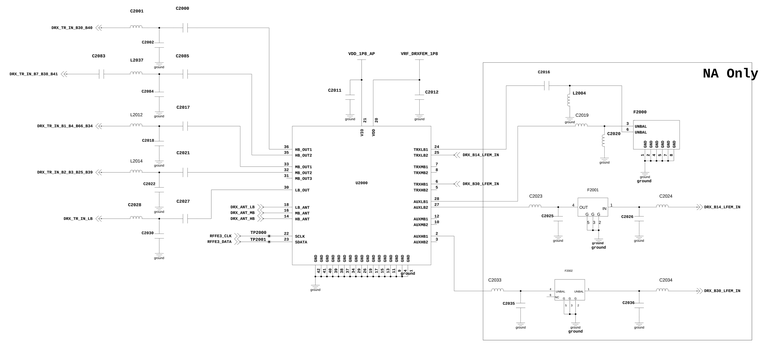
Rear CONTACT

Front CONTACT

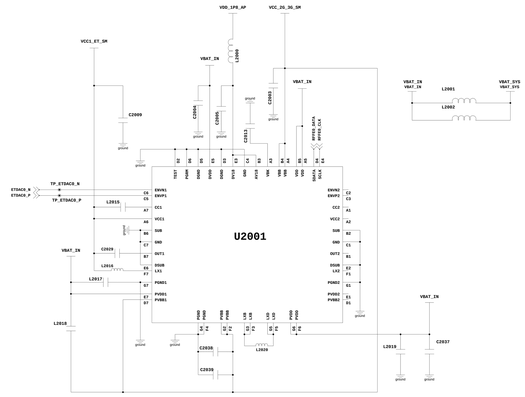
SD/SIM FPCB CONTACT

TCXO SHIELDCAN

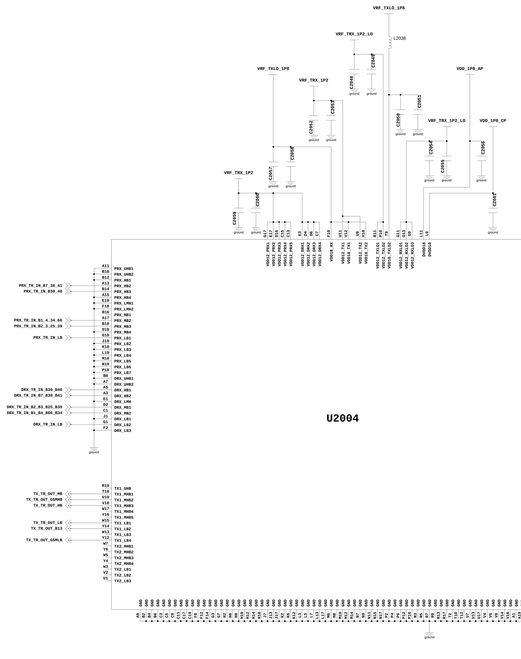
DRX_FEM



ET MODULATOR



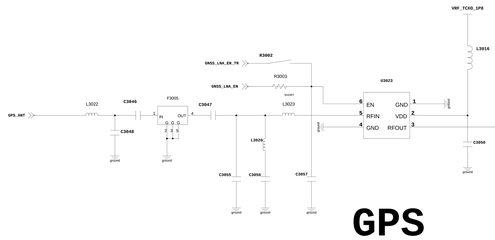
TRANCEIVER



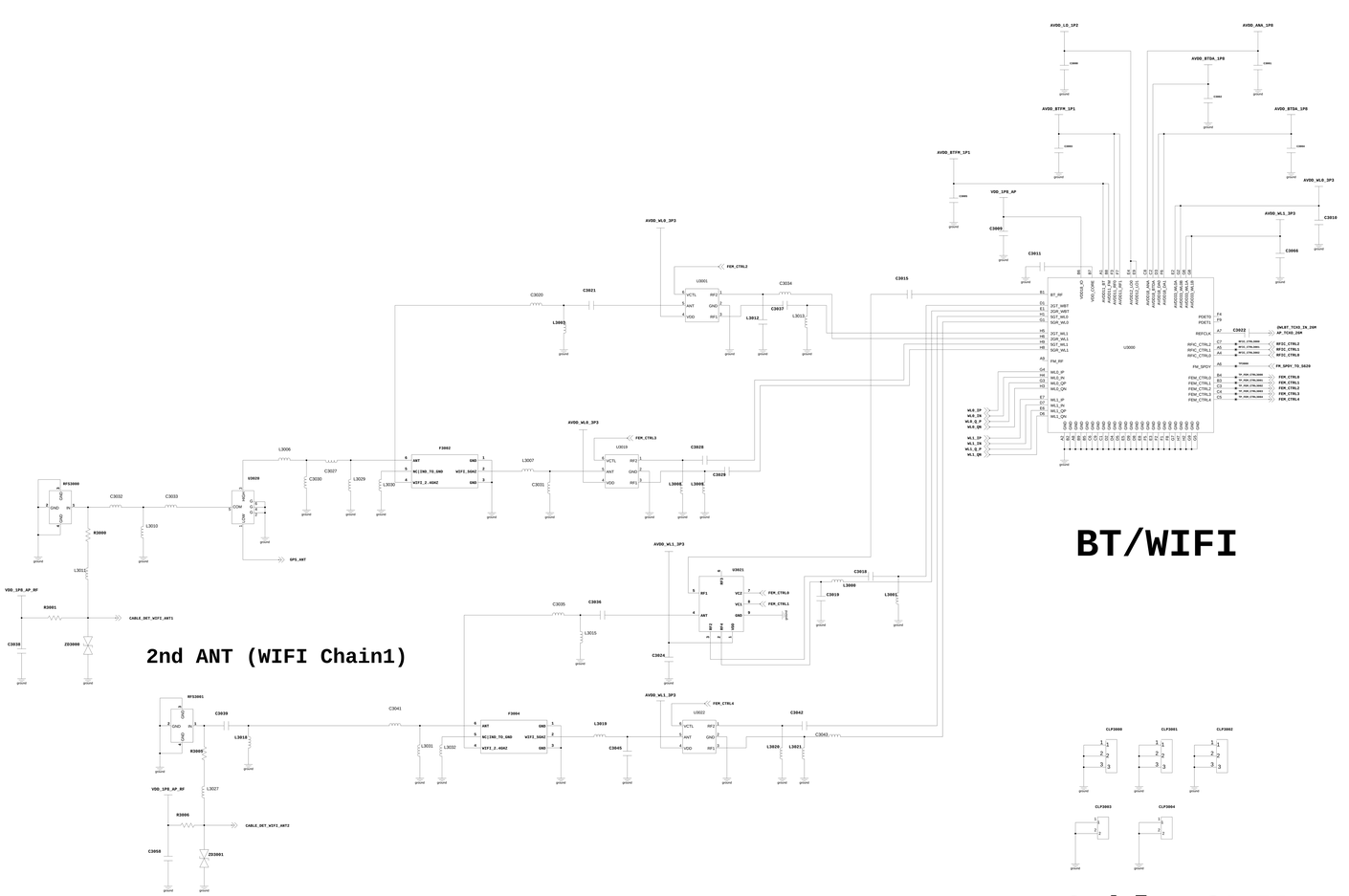
CONNECTIVITY

WIFI ONLY

GPS RECEIVER

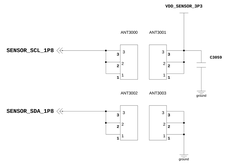


GPS

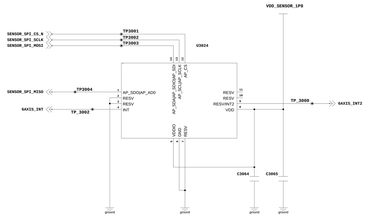


BT/WIFI

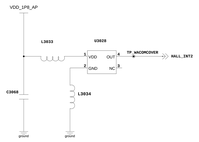
SENSOR



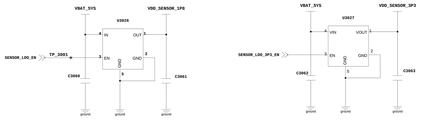
RGB SENSOR



6 AXIS(ACC+GRYO)



HALL_IC



SENSOR LDO

Project	10_SM_A505GT_LA_4_64_REV0		
Size	DWG NUMBER		Rev
A1			
Author	User Name	Sheet	4 of 8
		Date	Drawing Date

EXYNOS9611(RAMEN)_1

POP MEMORY

UCP4000

JTAG

BOOT

PMU

CLOCK

OTP

ADC

UART

SPI

I2C

(1/5)

I2S

CAM

CMGP

SHUB

PWM/TE

EINT

GPIO

UCP4000

CP GPIO

CP UART

ANT-SW LNA CTRL

USIM

RFIC CTRL

BT/WIFI/FM

(2/5)

MIPI RFFE

RX ADC0

RX ADC1

RX ADC2

TX DAC0

TX DAC1

ET DAC

GNSS

UCP4000

(3/5)

I2C Pull-up Register

PCB JTAG

AP JTAG

BLANKET THEM

H/W Revision Setting

REV	HW_VER3	HW_VER2	HW_VER1	HW_VER0
BRING-UP_B	0	0	0	0
BRING-UP_10	0	0	0	1
0.0	0	0	1	0
0.1	0	0	1	1
0.2	0	1	0	0
0.3	0	1	0	1
0.4	0	1	1	0
0.4A	0	1	1	1

BOOT option (need to update for RAME

OM[2:0]	1st	2nd
000	UFS	USB
001	USB	-
010	SD CARD	USB
011	UFS	SD CARD

Tab_A4_XL_Bringup

A1 — PDM NUMBER — Rev

User Name — SHEET# 8 — Drawing Date

EXYNOS9611(RAMEN)_2

UCP4000

USB 2.0

USB_AP_D_N — XUSB20_DM (AF1)
USB_AP_D_P — XUSB20_DP (AE1)
RS000 (ground) — XUSB20_TXRTUNE (AE2)
VDD_USB20_0P85 — C5000 (ground) — VDD085_USB20_PHY (Y6)
AVDD_D_P_1P8 — C5001 (ground) — AVDD18_USB20_PHY (AC7)
AVDD_USB20_PHY_3P3 — C5002 (ground) — AVDD33_USB20_PHY (AB5)

USB 3.0

XUSB0P_SS_RX1_N (AB2)
XUSB0P_SS_RX1_P (AB1)
XUSB0P_SS_TX1_N (W2)
XUSB0P_SS_TX1_P (W1)
XUSB0P_REXT1 (W3)
XUSB0P_SS_RX2_N (Y2)
XUSB0P_SS_RX2_P (Y1)
XUSB0P_SS_TX2_N (AC2)
XUSB0P_SS_TX2_P (AC1)
XUSB0P_REXT2 (AA3)
XDP_AUX_N (AA2)
XDP_AUX_P (AA1)
XDP_REXT (AA5)
AVDD085_DP (AA6)
VDD085_USB30_PHY1 (W6)
VDD085_USB30_PHY2 (V6)
AVDD_D_P_1P8 — AVDD18_DP (AC6)
AVDD033_USB30_PHY1 (AA4)
AVDD033_USB30_PHY2 (AB4)

MSH

XMMC_EMBD_D0 (A20)
XMMC_EMBD_D1 (A21)
XMMC_EMBD_D2 (G18)
XMMC_EMBD_D3 (B21)
XMMC_EMBD_D4 (C19)
XMMC_EMBD_D5 (F19)
XMMC_EMBD_D6 (G19)
XMMC_EMBD_D7 (B20)
XMMC_EMBD_RDQS (D18)
XMMC_EMBD_CMD (C18)

C5022 (ground), C5033 (ground), L5000, L5001

T_FLASH_D0 — XMMC_CARD_D0 (E16)
T_FLASH_D1 — XMMC_CARD_D1 (F15)
T_FLASH_D2 — XMMC_CARD_D2 (E17)
T_FLASH_D3 — XMMC_CARD_D3 (F16)
T_FLASH_CMD — R5004 — XMMC_CARD_CMD (F17)
T_FLASH_CLK — XMMC_CARD_CLK (G16)
VDD_1P8_AP — VDD_1P8_AP — VDDP18_MMC_CARD (J22), VDDQ18_MMC_EMBD (H24)
VDDQ_MMC_CARD_1P8_3P3 — VDDQ1833_MMC_CARD (H21)
C5011 (ground), C5012 (ground)
UFS_REFCLK — XMMC_EMBD_CLK (F18)
UFS_RST_N — XMMC_EMBD_nRESET (E18)

UFS

UFS_RX_D0_N — TP_UFS_RX_N (A23)
UFS_RX_D0_P — TP_UFS_RX_P (B23)
UFS_TX_D0_N — XUFS_EMBD_TXD_N (A24)
UFS_TX_D0_P — XUFS_EMBD_TXD_P (B24)
RS003 (ground) — XUFS_EMBD_REXT (A25)
VDD_PLL_AUD_0P85 — AVDD085_UFS_EMBD (J25)
AVDD_MIPI_DPHY_1P2 — VDDQ1218_UFS_EMBD_MMC (H23)
AVDD_PLL_CPUCL0_1P8 — AVDD18_UFS_EMBD (J24), AVDD18_CMN_UFS_EMBD (K23)
AVDD_PLL_CPUCL0_1P8
C5018 (ground), C5019 (ground), C5020 (ground), C5021 (ground)

MIPI

(4/5)

XMIPI_DSI4L_D0_P (J1) — LCD_MIPI_D0_P
XMIPI_DSI4L_D0_N (H1) — LCD_MIPI_D0_N
XMIPI_DSI4L_D1_P (J2) — LCD_MIPI_D1_P
XMIPI_DSI4L_D1_N (H2) — LCD_MIPI_D1_N
XMIPI_DSI4L_D2_P (J4) — LCD_MIPI_D2_P
XMIPI_DSI4L_D2_N (H4) — LCD_MIPI_D2_N
XMIPI_DSI4L_D3_P (J5) — LCD_MIPI_D3_P
XMIPI_DSI4L_D3_N (H5) — LCD_MIPI_D3_N
XMIPI_DSI4L_CLK_P (J3) — LCD_MIPI_CLK_P
XMIPI_DSI4L_CLK_N (H3) — LCD_MIPI_CLK_N

XMIPI_DSI2L_D0_P (F5)
XMIPI_DSI2L_D0_N (F4)
XMIPI_DSI2L_D1_P (F3)
XMIPI_DSI2L_D1_N (F2)
XMIPI_DSI2L_CLK_P (E2)
XMIPI_DSI2L_CLK_N (E1)

XMIPI_DSI2L_VREG_0P4V (N5)
XMIPI_DSI4L_VREG_0P4V (N4) — C5010 — ground

XMIPI_CSI4LA_D0_P (M3) — RCAM_MIPI_D0_P
XMIPI_CSI4LA_D0_N (M2) — RCAM_MIPI_D0_N
XMIPI_CSI4LA_D1_P (L4) — RCAM_MIPI_D1_P
XMIPI_CSI4LA_D1_N (K4) — RCAM_MIPI_D1_N
XMIPI_CSI4LA_D2_P (L2) — RCAM_MIPI_D2_P
XMIPI_CSI4LA_D2_N (K2) — RCAM_MIPI_D2_N
XMIPI_CSI4LA_D3_P (L3) — RCAM_MIPI_D3_P
XMIPI_CSI4LA_D3_N (K3) — RCAM_MIPI_D3_N
XMIPI_CSI4LA_CLK_P (M5) — RCAM_MIPI_CLK_P
XMIPI_CSI4LA_CLK_N (M4) — RCAM_MIPI_CLK_N

XMIPI_CSI4LB_D0_P (U2) — FCAM_MIPI_D0_P
XMIPI_CSI4LB_D0_N (U1) — FCAM_MIPI_D0_N
XMIPI_CSI4LB_D1_P (T2) — FCAM_MIPI_D1_P
XMIPI_CSI4LB_D1_N (T1) — FCAM_MIPI_D1_N
XMIPI_CSI4LB_D2_P (P2)
XMIPI_CSI4LB_D2_N (P1)
XMIPI_CSI4LB_D3_P (N2)
XMIPI_CSI4LB_D3_N (N1) — ground
XMIPI_CSI4LB_CLK_P (R2) — FCAM_MIPI_CLK_P
XMIPI_CSI4LB_CLK_N (R1) — FCAM_MIPI_CLK_N

XMIPI_CSI4LC_D0_P (C3)
XMIPI_CSI4LC_D0_N (D3)
XMIPI_CSI4LC_D1_P (A4)
XMIPI_CSI4LC_D1_N (B4)
XMIPI_CSI4LC_D2_P (A5)
XMIPI_CSI4LC_D2_N (B5)
XMIPI_CSI4LC_D3_P (C5)
XMIPI_CSI4LC_D3_N (D5)
XMIPI_CSI4LC_CLK_P (C4)
XMIPI_CSI4LC_CLK_N (D4) — ground

XMIPI_CSI4LD_D0_P (A7)
XMIPI_CSI4LD_D0_N (A8)
XMIPI_CSI4LD_D1_P (B7)
XMIPI_CSI4LD_D1_N (B8)
XMIPI_CSI4LD_D2_P (D7)
XMIPI_CSI4LD_D2_N (D8)
XMIPI_CSI4LD_D3_P (E7)
XMIPI_CSI4LD_D3_N (E8) — ground
XMIPI_CSI4LD_CLK_P (C7)
XMIPI_CSI4LD_CLK_N (C8)

XMIPI_CSI2L_D0_P (A10)
XMIPI_CSI2L_D0_N (B10)
XMIPI_CSI2L_D1_P (C10)
XMIPI_CSI2L_D1_N (D10) — ground
XMIPI_CSI2L_CLK_P (C11)
XMIPI_CSI2L_CLK_N (D11)

AVDD085_MIPI_DPHY (K6)
AVDD085_MIPI_DPHY (L6)
AVDD085_MIPI_DPHY (R7)
AVDD085_MIPI_DPHY (T7) — VDD_PLL_AUD_0P85
C5013 (ground), C5014 (ground)

AVDD12_MIPI_DPHY (P5) — AVDD_MIPI_DPHY_1P2
AVDD18_MIPI_DPHY (N6) — AVDD_PLL_CPUCL0_1P8
C5015 (ground), C5016 (ground), C5017 (ground)

UCP4000

(5/5)

VSS: A1, B1, C1, V1, AD1, AG1, AJ1, AK1, AL1, A2, B2, C2, G2, V2, AD2, AH2, AK2, AL2, A3, E3, G3, N3, P3, R3, T3, U3, Y3, AB3, AC3, AD3, AL3, E4, G4, P4, T4, E5, G5, K5, L5, Y5, AL5, A6, D6, E6, F6, J6, T6, AE6, F7, J7, K7, L7, M7, AB7, F8, J8, AL8, A9, D9, E9, K9, L9, N9, P9, R9, T9, U9, W9, Y9, AA9, AB9, AE9, AK9, E10, H10, E11, L11, N11, A12, B12, C12, D12, G12, T12, V12, G13, AA13, G14, T14, V14, AC14, AL14, B15, J15, K15, L15, N15, AA15, AD15, H16, C17, R17, U17, W17, AA17, A18, AK18, D19, E19, H19, L19, R19, U19, AA19, AD19, AL19, C20, F20, G20, H20, L21, N21, R21, T21, U21, V21, W21, Y21, AA21, AL21, A22, B22, C22, H22, V22, Y22, C23, J23, R23, T23, W23, Y23, AA23, AC23, AJ23, B24, E24, K24, AC24, F25, G25, AL25, A26, N26, R26, T26, AA26, AD26, A27, B27, G27, H27, K27, R27, AA27, AD27, AK27, AL27, N28, B28, C28, P28, AD28, AK28, AL28, A29, B29, C29, F29, H29, P29, AD29, AJ29, AK29, AL29, N24, V26, Y26, U27, Y27

UFS MEMORY

VDD_UFS_2P95, VDD_UFS_1P8

C5003 (ground), C5004 (ground), C5005 (ground), C5006 (ground), C5007 (ground), C5008 (ground), C5009 (ground)

U5000

TP5004 — VSF3 (E9)
TP5000 — VSF4 (E10)
TP5005 — VSF5 (F10)
VSF6 (G5)
TP5001 — VSF7 (J10)
VSF8 (K10)
VSF9 (P10)

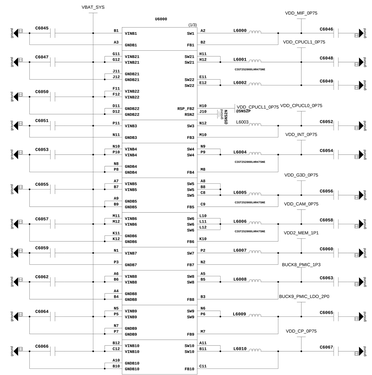
UFS_REFCLK — TP5002 — REF_CLK (H1)
UFS_RST_N — TP5003 — RTS_N (H2)
UFS_TX_D0_N — TP_UFS_TX_N — DIN0_C (F2)
UFS_TX_D0_P — TP_UFS_TX_P — DIN0_T (F1)
UFS_RX_D0_N — DOUT0_C (K1)
UFS_RX_D0_P — DOUT0_T (K2)

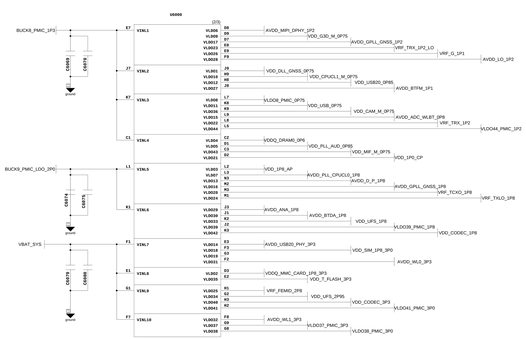
NC: A1, A2, B1, B4, B5, M1, M8, M9, N1, N14, P1, P2, P13, P14, B3
VCCQ: A4, B10, B14, C4, C5
VCCQ2: D1, D2, E5, F5, K9, M2, M11, N2
CPOUT1: A10
CPOUT2: B10
C-: A11
C+: A12
DIN1_C: D4
INDEXINC: E5
DOUT1_C: M1
DOUT1_T: M2
RFU: B13, C10, C13, C14, E13, E14, G13, G14
VCCQ: A5, B14, C5
VCCQ2: D1, D2, F5, K9, M11, N1
VSS: A3, A6, A7, A8, A9, B6, B7, B8, B9, C6, C7, C8, C9, D6, D7, D8, D9, E6, E7, E8, E9, F6, F7, F8, F9, G6, G7, G8, G9, H6, H7, H8, H9, J6, J7, J8, J9, K6, K7, K8, L6, L7, L8, L9, M6, M7, M8, M9, N6, N7, N8, N9, P6, P7, P8, P9, R6, R7, R8, R9, T6, T7, T8, T9, U6, U7, U8, U9, V6, V7, V8, V9, W6, W7, W8, W9, Y6, Y7, Y8, Y9, AA6, AA7, AA8, AA9, AB6, AB7, AB8, AB9, AC6, AC7, AC8, AC9, AD6, AD7, AD8, AD9, AE6, AE7, AE8, AE9, AF6, AF7, AF8, AF9, AG6, AG7, AG8, AG9, AH6, AH7, AH8, AH9, AJ6, AJ7, AJ8, AJ9, AK6, AK7, AK8, AK9, AL6, AL7, AL8, AL9, AM6, AM7, AM8, AM9
VDD1: A9
VCC: B6, B7, B8, B9, C6, C7, C8, C9, D6, D7, D8, D9, E6, E7, E8, E9, F6, F7, F8, F9, G6, G7, G8, G9, H6, H7, H8, H9, J6, J7, J8, J9, K6, K7, K8, L6, L7, L8, L9, M6, M7, M8, M9, N6, N7, N8, N9, P6, P7, P8, P9, R6, R7, R8, R9, T6, T7, T8, T9, U6, U7, U8, U9, V6, V7, V8, V9, W6, W7, W8, W9, Y6, Y7, Y8, Y9, AA6, AA7, AA8, AA9, AB6, AB7, AB8, AB9, AC6, AC7, AC8, AC9, AD6, AD7, AD8, AD9, AE6, AE7, AE8, AE9, AF6, AF7, AF8, AF9, AG6, AG7, AG8, AG9, AH6, AH7, AH8, AH9, AJ6, AJ7, AJ8, AJ9, AK6, AK7, AK8, AK9, AL6, AL7, AL8, AL9, AM6, AM7, AM8, AM9
VCCQ: A4, B10, B14, C4, C5
VCCQ2: D1, D2, E5, F5, K9, M2, M11, N2
VDD2: A9
RFU: J13, J14, L13, L14, M6, M7, M10, N6, N7, N10, N13, P3, P6, P7
VSS: M3, G10, M5, D14, D13, M4, B2, E1, M12, M13, M14, L3, N2, N3, N4, N5, D3, C12, C13, E12, N11, N12, J3, J2, J1, H14, C3, P4, P5, C2, C1, F12, B12, F14, P11, P12, G3, B11

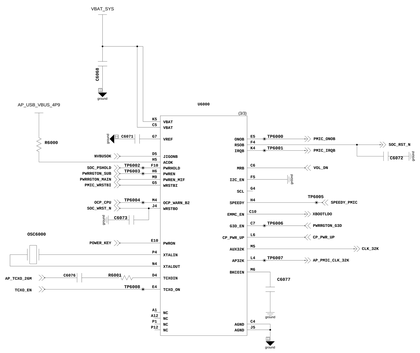
TITLE: 10_SM_A505GT_LA_4_64_REV0.1
SIZE: A1 — DWG NO: PDM NUMBER — Rev
DRAWN BY: User Name — SHEET: 10 of 10 — Date: Drawing Date

POWER

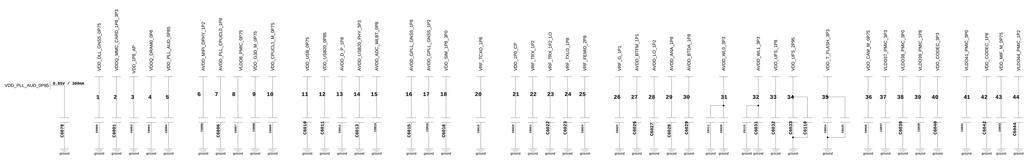
MAIN_PMIC





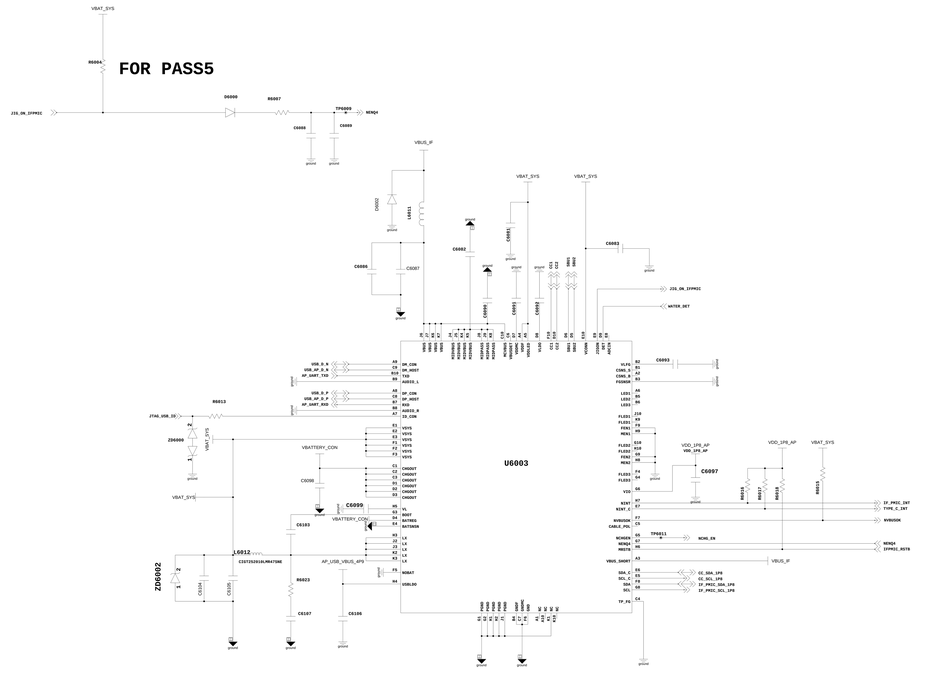


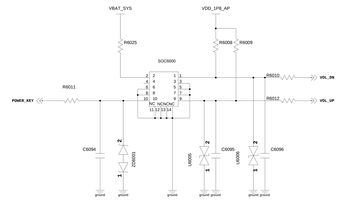
SYSTEM POWER



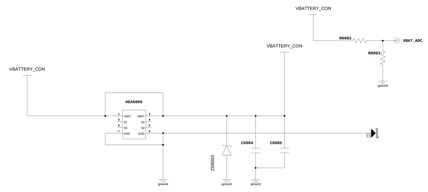
IF_PMIC SM5713

FOR PASS5

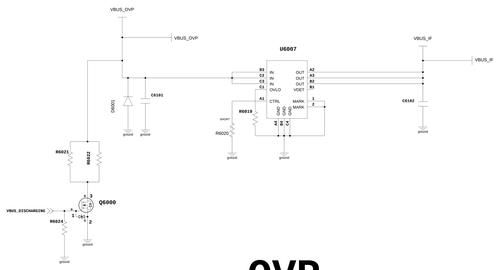




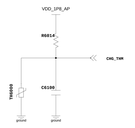
KEY CONN



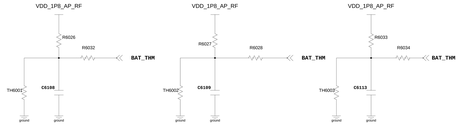
BATTERY CON



OVP



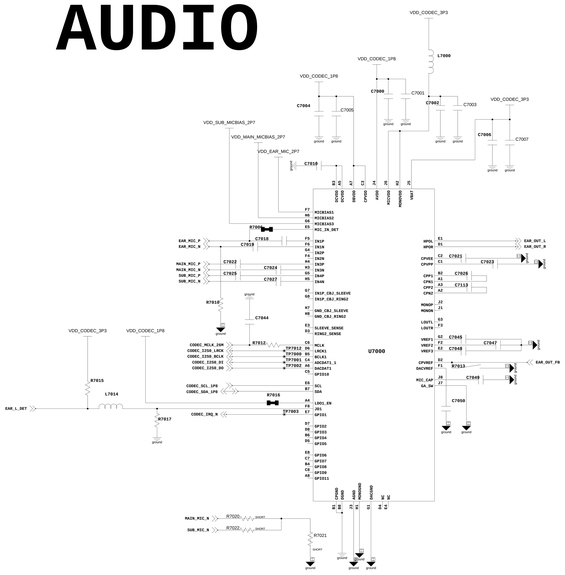
CHG THM



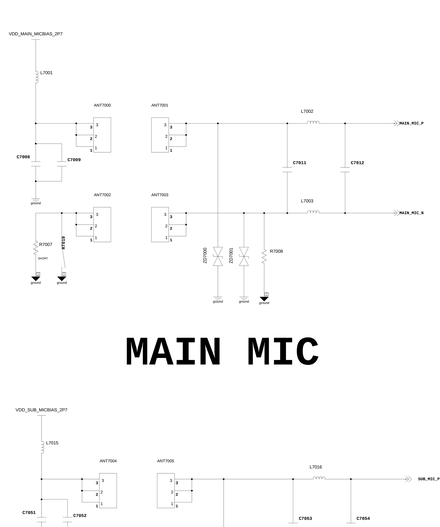
BAT THM_Option

TITLE	10_SM_A505GT_LA_4_64_REV0		
SIZE	A1	NUMBER	REV
ENGINEER	User Name	SHEET	6 OF 8
DATE	Drawing Date		

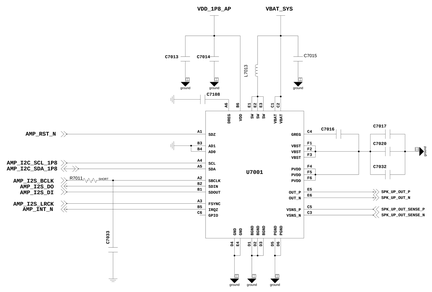
AUDIO



Audio CODEC



MAIN MIC



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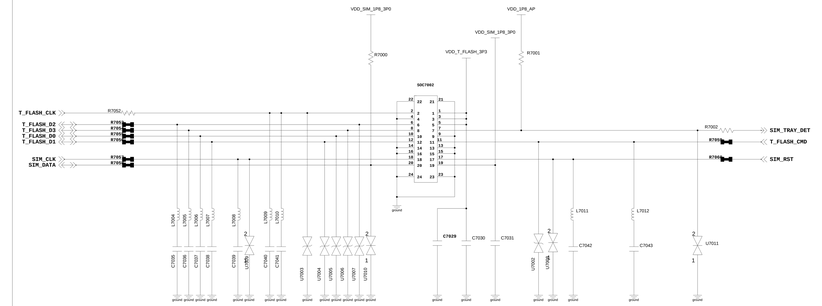
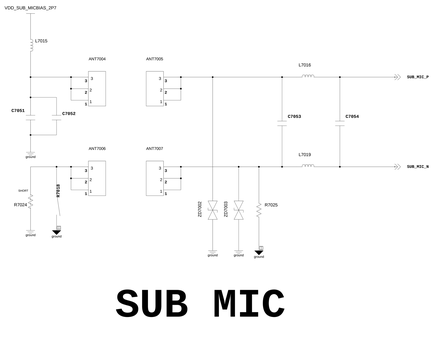
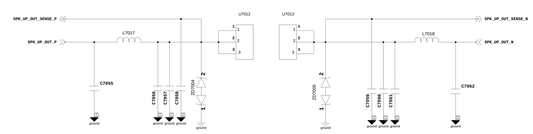


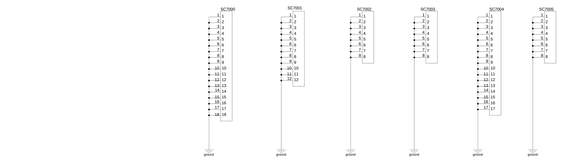
SIM & SD CON



SUB MIC

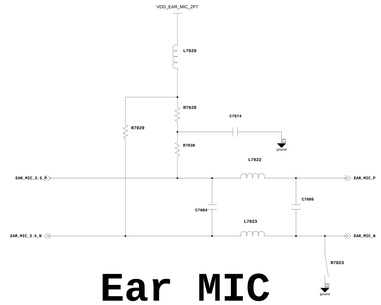


SPK UP CONTACT



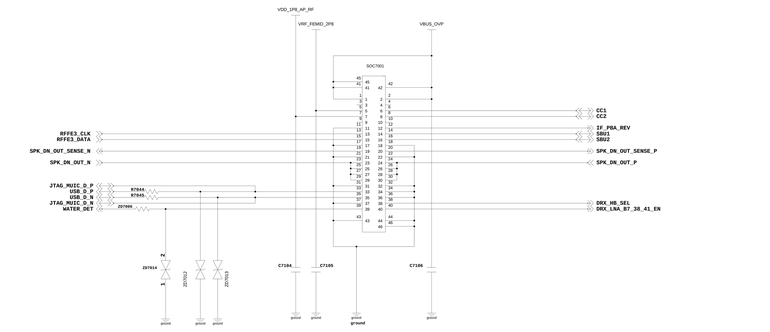
SIELD_CAN

WAKE UP IC



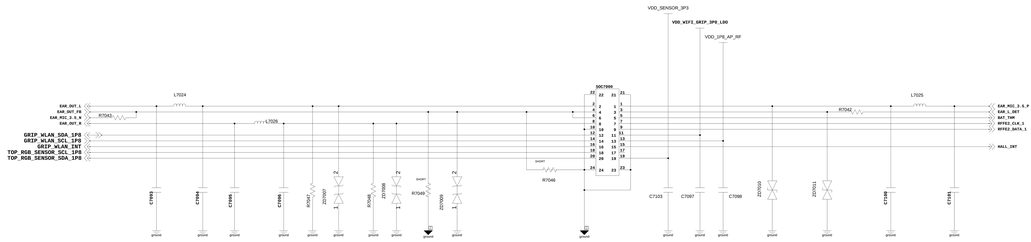
Ear MIC

SPK DOWN AMP



IF Connector

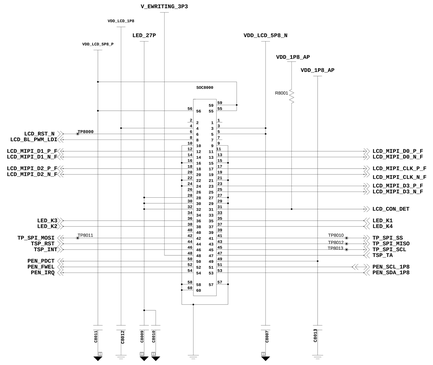
3.5pi Connector



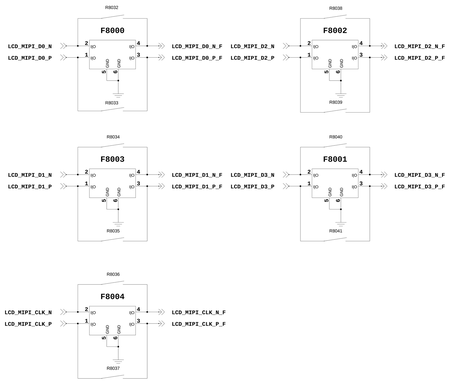
2D BARCODE

	EUR	NA	WIFI	ETC	64GB	128GB
R7034	Solder					
R7036		Solder				
R7039			Solder			
R7040				Solder		
R7061					Solder	
R7062						Solder

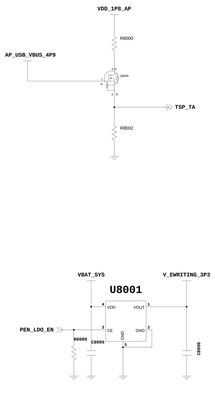
DISPLAY



LCD CON



LCD FILTER

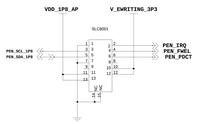


S-PEN LDO

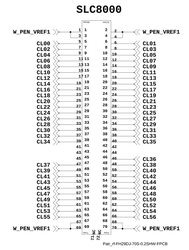
WACOM PCB

< SMD F/T >

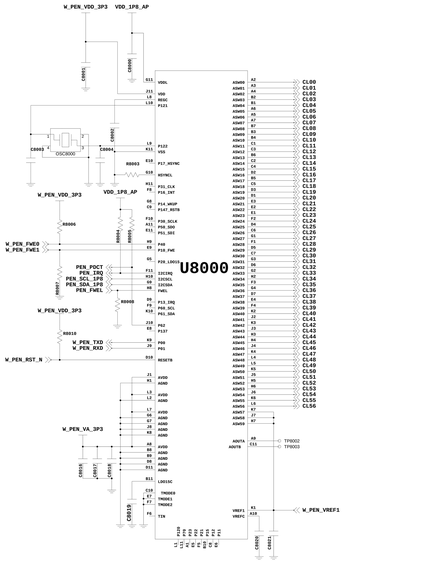
1. V_EWRITING_3P3
2. VDD_1P8_AP
3. PEN_SCL_1P8
4. PEN_SDA_1P8
5. PEN_FWEL
6. PEN_PDCT
7. GND
8. PEN_IRQ



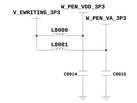
WACOM I/F CON

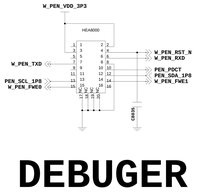


WACOM CON



S_PEN Control IC





DEBUGER

LCD BACKLIGHT DRIVER IC

LCD BUCK BOOSTER

LCD LDO

CAMERA

8M CAM(R_CAM)

R_CAM LDOS

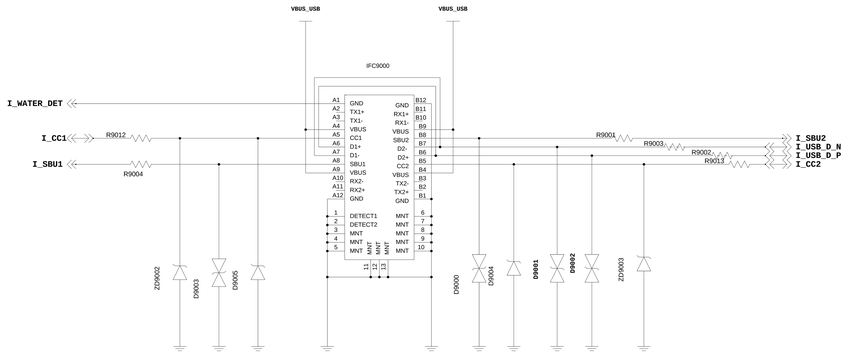
5M CAM (F_CAM)

F_CAM LDOS

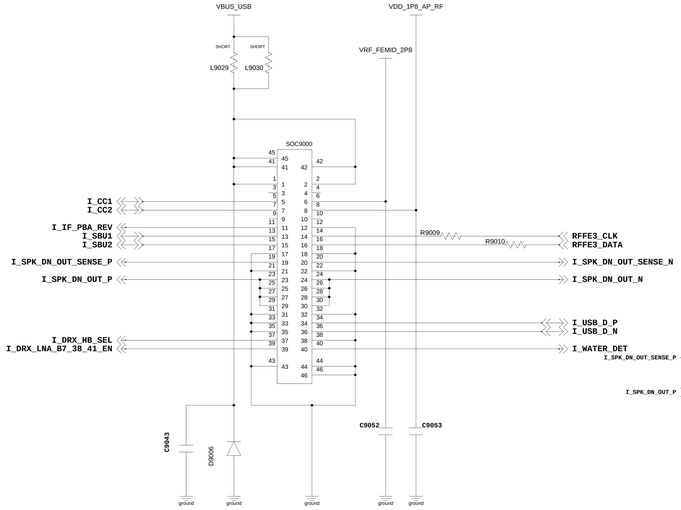
R_CAM Filter

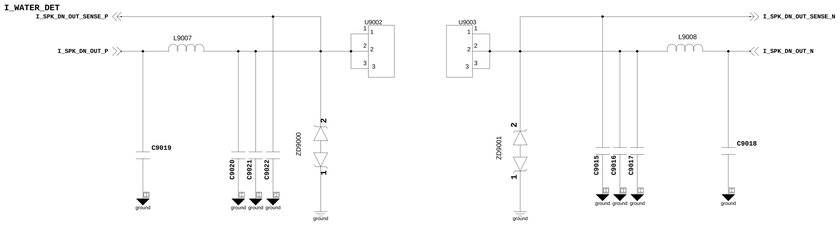
TITLE	10_SM_A505GT_LA_4_64_REV0			
SIZE	A1	DOCUMENT NUMBER		REV
DRAWN BY	User Name	SHEET	B of B	Drawing Date

IF PCB



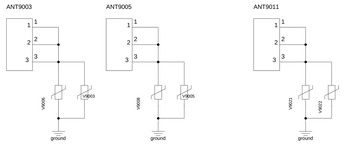
USB C-TYPE CONN





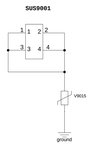
IF Con.

SPK DOWN CONTACT

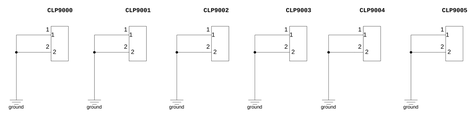


1.3T 0.9T

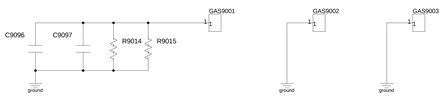
REAR CONTACT



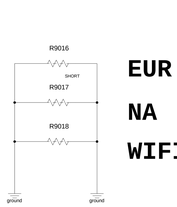
SCREW SUS(0.3T)



Cable CLIP



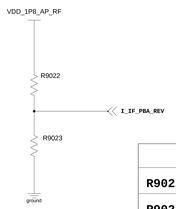
Front CONTACT



EUR

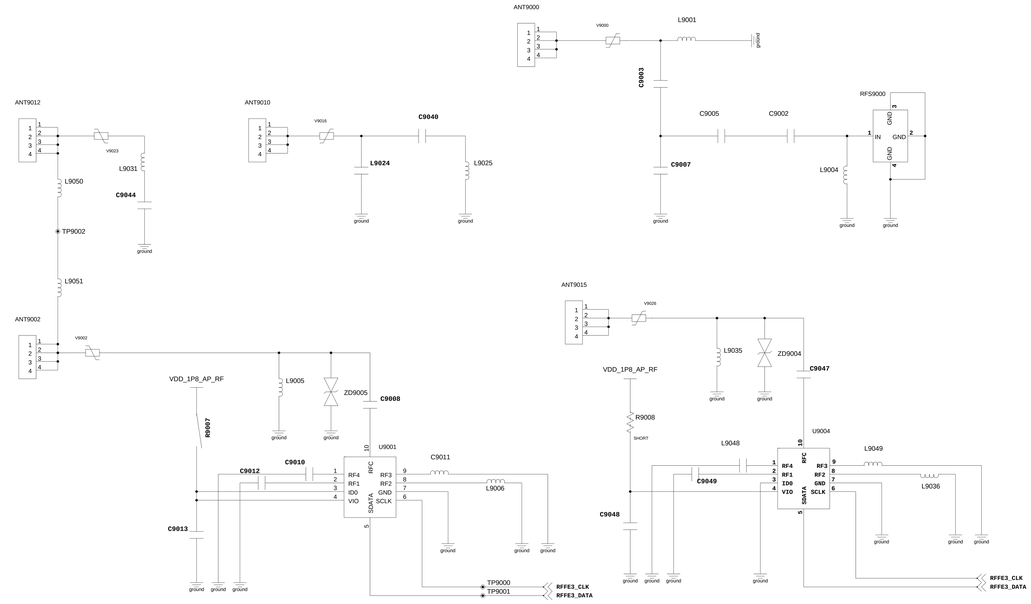
NA

WIFI

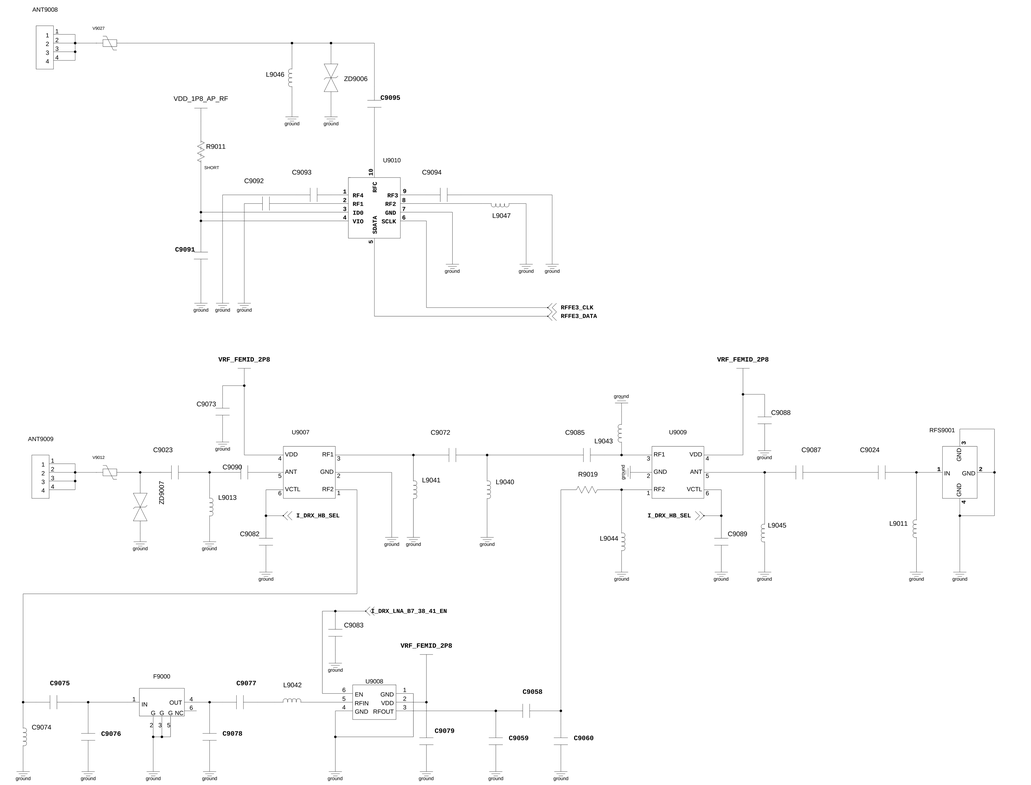


	EUR	NA	WIFI
R9022	OPEN	100K	100K
R9023	100K	OPEN	100K

SUB ANT1(LM)

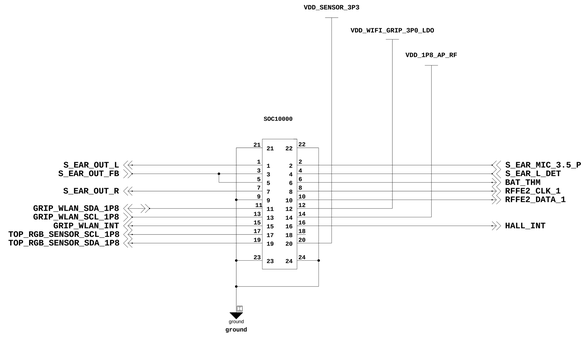


SUB ANT2(HB)

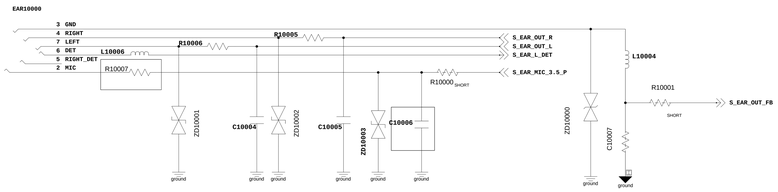


TITLE	10_SM_A505GT_LA_4_64_REV04C_		
Size A1	DWG NO PDM NUMBER		Rev
DRAWN BY User Name	SHEET 17 OF 9	Date Drawing Date	

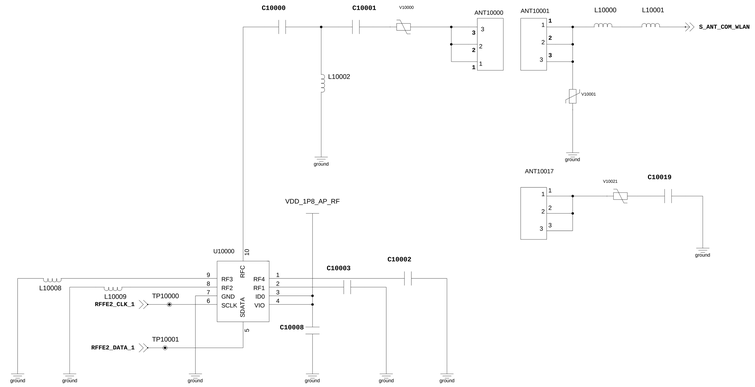
3.5pi PCB



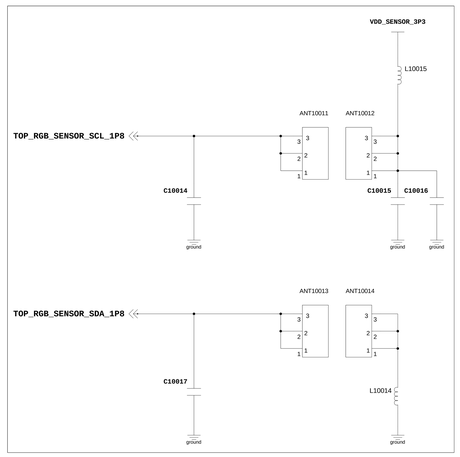
WIFI IF Con.



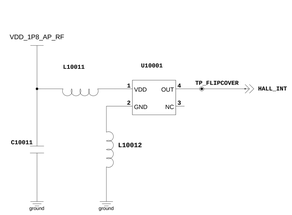
EARJACK



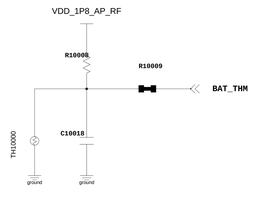
WIFI_1 ANT



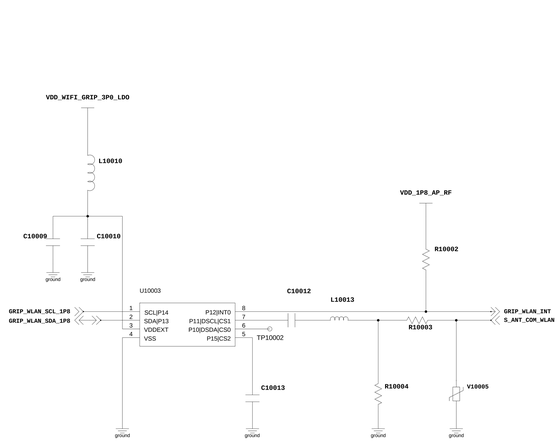
RGB SENSOR CON



HALL_IC



BAT_THM



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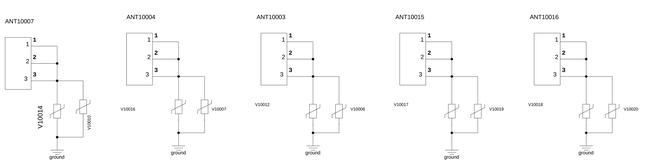
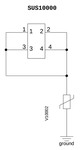


REAR CONTACT

Front CONTACT



Screw SUS

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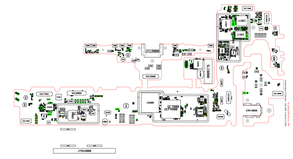

Bed
Bed
Bed